

MOSFET Amplifier Simulation Report

1. Introduction & Objective

- **Project Overview:** This project implements a comprehensive Python-based simulation framework for a Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) amplifier.
- **Objective:** To model core MOSFET physics, analyze DC operating points (Q-point), calculate small-signal parameters (g_m , voltage gain), and plot characteristic I-V curves.

2. Theoretical Background & Equations

- **Triode (Linear) Region ($V_{DS} < V_{GS} - V_{th}$):** The drain current is dependent on both V_{GS} and V_{DS} , acting like a voltage-controlled resistor.
- **Saturation Region ($V_{DS} \geq V_{GS} - V_{th}$):** The drain current becomes relatively independent of V_{DS} and is primarily controlled by the gate-source overdrive voltage.
- **Transconductance (g_m):** Determines the amplification capability, representing the change in drain current with respect to gate-source voltage.

3. Simulation Framework & Modules

- `mosfet_model.py` : Handles core device equations and region-switching logic.
- `biasing.py` & `gain.py` : Computes stable DC biasing points and AC voltage gain characteristics.
- `iv_curve.py` : Generates automated plots saved directly into the `graphs/` directory.

4. Conclusion

The simulation successfully models the non-linear behavior of MOSFETs and provides accurate analytical curves for amplifier design validation.